

# Topics in Category Theory

Labix

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**Abstract**

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# 1 Categorical Algebra

## 1.1 Monoid Objects

**Definition 1.1.1 (Monoid Objects)** Let  $(\mathcal{C}, \otimes, I)$  be a symmetric monoidal category. We say that  $M \in \mathcal{C}$  is a monoid object if there exists

- A morphism called multiplication  $\mu : M \otimes M \rightarrow M$
- A morphism called the unit  $\eta : I \rightarrow M$

such that the following coherence conditions hold.

- Associativity:

$$\begin{array}{ccc}
 (M \otimes M) \otimes M & \xrightarrow{\mu \otimes \text{id}_M} & M \otimes M \\
 \downarrow \cong & & \downarrow \mu \\
 M \otimes (M \otimes M) & \xrightarrow{\text{id}_M \otimes \mu} & M \otimes M \\
 & & \uparrow \mu \\
 & & M
 \end{array}$$

- Identity:

$$\begin{array}{ccccc}
 I \otimes M & \xrightarrow{\eta \otimes \text{id}_M} & M \otimes M & \xleftarrow{\text{id}_M \otimes \mu} & M \otimes I \\
 & \searrow \cong & \downarrow \mu & \swarrow \cong & \\
 & & M & & 
 \end{array}$$

**Proposition 1.1.2** A monoid object in  $(\mathbf{Set}, \times, *)$  is a monoid.

There is an interesting categorization that can be seen here from the traditional algebraic structures. Recall that a monoid is a set  $M$  together with a binary operation  $+$  :  $M \times M \rightarrow M$  such that addition is associativity, and that there exists an identity element  $e$  for which for any  $m \in M$ , we have  $e + m = m = m + e$ . Reorganizing the data, we can write this as the following: A monoid is a set together with

- A multiplication map  $+$  :  $M \times M \rightarrow M$
- An unit map  $\eta : * \rightarrow M$

such that the following conditions are satisfied:

- The multiplication map is associative
- The unit map is such that multiplication with the unit returns itself.

This is starting point of formalizing monoid objects in category theory. An algebraic object in general consists of a set, some binary operations together with some coherence conditions for the binary operations. The coherence conditions are in particular provided by the associator and unitor of the underlying symmetric monoidal structure. One can in particular think of the unit map  $\eta : * \rightarrow M$  as picking out a point in  $M$  to be the identity in  $M$ .

It is important to make clear what are the data of an object and what are the conditions the data are subject to. In our case, the data consists a set, a binary operation and an identity element. These data are subject to conditions called associativity and unity which is dependent on the symmetric monoidal structure.

## 1.2 Group Objects

**Definition 1.2.1 (Group Objects)** Let  $(\mathcal{C}, \otimes, I)$  be a symmetric monoidal category. We say that  $G \in \mathcal{C}$  is a group object if there exists

- A morphism called multiplication  $\mu : G \otimes G \rightarrow G$
- A morphism called the unit  $\eta : I \rightarrow M$
- A morphism called the inverse  $\text{inv} : G \rightarrow G$

such that the following coherence conditions hold.

- Associativity:

$$\begin{array}{ccc}
 (G \otimes G) \otimes G & \xrightarrow{\mu \otimes \text{id}_G} & G \otimes G \\
 \downarrow \cong & & \downarrow \mu \\
 & & G \\
 & & \uparrow \mu \\
 G \otimes (G \otimes G) & \xrightarrow{\text{id}_G \otimes \mu} & G \otimes G
 \end{array}$$

- Identity:

$$\begin{array}{ccccc}
 I \otimes G & \xrightarrow{\eta \otimes \text{id}_G} & G \otimes G & \xleftarrow{\text{id}_G \otimes \mu} & G \otimes I \\
 & \searrow \cong & \downarrow \mu & & \swarrow \cong \\
 & & M & & 
 \end{array}$$

- Inverse:

$$\begin{array}{ccc}
 G & \xrightarrow{(\text{id}_G, \text{inv})} & G \otimes G \\
 (\text{inv}, \text{id}_G) \downarrow & \searrow \text{id}_G & \downarrow \mu \\
 G \otimes G & \xrightarrow{\mu} & G
 \end{array}$$

**Lemma 1.2.2** Let  $\mathcal{C}$  be a symmetric monoidal category. Then every group object in  $\mathcal{C}$  is also a monoid object.

**Theorem 1.2.3** The following are group objects in particular categories.

- The group objects in  $(\mathbf{Set}, \times, *)$  are groups
- The group objects in  $(\mathbf{Grp}, \times, *)$  are abelian groups
- The group objects in  $(\mathbf{Ab}, \oplus, *)$  are abelian groups
- The group objects in  $(\mathbf{Top}, \times, *)$  are topological groups

### 1.3 Ring Objects

### 1.4 Algebra and Coalgebra Objects

There is a need to revisit the definition of an algebra (over a field)

**Proposition 1.4.1** A vector space  $V$  over a field  $k$  is an algebra if and only if there is a following collection of data:

- A  $k$ -linear map  $m : V \otimes V \rightarrow V$  called the multiplication map
- An  $k$ -linear map  $u : k \rightarrow V$  called the unital map

such that the following two diagrams are commutative:

$$\begin{array}{ccc}
 V \otimes V \otimes V & \xrightarrow{\text{id} \otimes m} & V \otimes V \\
 m \otimes \text{id} \downarrow & & \downarrow m \\
 V \otimes V & \xrightarrow{m} & V
 \end{array}
 \qquad
 \begin{array}{ccc}
 k \otimes V & \xrightarrow{u \otimes \text{id}} & V \otimes V \\
 \cong \downarrow & \swarrow m & \uparrow \text{id} \otimes u \\
 V & \xleftarrow{\cong} & V \otimes k
 \end{array}$$

where the unnamed maps are the canonical isomorphisms.

Evidently, the map  $\mu$  gives a multiplicative structure for  $V$  and  $\Delta$  gives the unitary structure of an algebra. The diagram on the left then represents associativity of multiplication. Notice that such additional structure on  $V$  formally lives in the category  $\mathbf{Vect}_k$  of vector spaces over a fixed field  $k$ .

Therefore we can formally dualize all arrows to obtain a new object.

**Definition 1.4.2 (Coalgebra)** Let  $V$  be a vector space over a field  $k$ . We say that  $V$  is a coalgebra over  $k$  if there is a collection of data:

- A  $k$ -linear map  $\Delta : V \rightarrow V \otimes V$  called the comultiplication map
- An  $k$ -linear map  $\varepsilon : V \rightarrow k$  called the counital map

such that the following diagrams are commutative:

$$\begin{array}{ccc} V \otimes V \otimes V & \xleftarrow{\varepsilon \otimes \Delta} & V \otimes V \\ \Delta \otimes \varepsilon \uparrow & & \uparrow \Delta \\ V \otimes V & \xleftarrow{\Delta} & V \end{array} \quad \begin{array}{ccc} k \otimes V & \xleftarrow{\varepsilon \otimes \text{id}} & V \otimes V \\ \cong \uparrow & \nearrow \Delta & \downarrow \text{id} \otimes \varepsilon \\ V & \xrightarrow{\cong} & V \otimes k \end{array}$$

where the unnamed maps are the canonical isomorphisms.

**Lemma 1.4.3** Every vector space  $V$  over a field  $k$  can be given the structure of a coalgebra where

- $\Delta : V \rightarrow V \otimes V$  is defined by  $\Delta(v) = v \otimes v$
- $\varepsilon : V \rightarrow k$  is defined by  $\varepsilon(v) = 1_k$

We would like to formally invert the definitions of algebra homomorphisms in order to define coalgebra homomorphisms.

## 1.5 Bialgebra Objects

**Definition 1.5.1 (Bialgebras)** Let  $V$  be a vector space over a field  $k$ . We say that  $V$  is a bialgebra if there is a collection of data:

- A  $k$ -linear map  $m : V \otimes V \rightarrow V$  called the multiplication map
- An  $k$ -linear map  $u : k \rightarrow V$  called the unital map
- A  $k$ -linear map  $\Delta : V \rightarrow V \otimes V$  called the comultiplication map
- An  $k$ -linear map  $\varepsilon : V \rightarrow k$  called the counital map

such that  $(V, m, u)$  is an algebra over  $k$  and  $(V, \Delta, \varepsilon)$  is a coalgebra over  $k$  and that the following diagrams are commutative:

$$\begin{array}{ccc} V \otimes V & \xrightarrow{m} & V \\ \Delta \otimes \Delta \downarrow & & \uparrow m \otimes m \\ V \otimes V \otimes V \otimes V & \xrightarrow{\text{id} \otimes \tau \otimes \text{id}} & V \otimes V \otimes V \otimes V \end{array} \quad \begin{array}{ccc} k & \xrightarrow{\text{id}} & k \\ u \searrow & & \nearrow \varepsilon \\ & V & \end{array}$$

$$\begin{array}{ccc} V \otimes V & \xrightarrow{m} & V \\ \varepsilon \otimes \varepsilon \searrow & & \swarrow \varepsilon \\ & k \otimes k \cong k & \end{array} \quad \begin{array}{ccc} & k \otimes k \cong k & \\ u \otimes u \swarrow & & \searrow u \\ V \otimes V & \xleftarrow{\Delta} & V \end{array}$$

where  $\tau : V \otimes V \rightarrow V \otimes V$  is the commutativity map defined by  $\tau(x \otimes y) = y \otimes x$ .

**Theorem 1.5.2** Let  $V$  be a vector space over  $k$ . Suppose that  $(V, m, u)$  is an algebra and  $(V, \Delta, \varepsilon)$  is a coalgebra. Then the following conditions are equivalent.

- $(V, m, u, \Delta, \varepsilon)$  is a bialgebra
- $m : V \otimes V \rightarrow V$  and  $u : k \rightarrow V$  are coalgebra homomorphisms
- $\Delta : V \rightarrow V \otimes V$  and  $\varepsilon : V \rightarrow k$  are algebra homomorphisms

## 1.6 Hopf Algebra Objects

**Definition 1.6.1 (Hopf Algebra)** Let  $(H, m, u, \Delta, \varepsilon)$  be a bialgebra. We say that  $H$  is a Hopf algebra if there is a  $k$ -linear map  $S : H \rightarrow H$  called the antipode such that the following diagram commutes:

$$\begin{array}{ccccc}
 & H \otimes H & \xrightarrow{S \otimes \text{id}} & H \otimes H & \\
 \Delta \nearrow & & & & \searrow m \\
 H & \xrightarrow{\varepsilon} & k & \xrightarrow{u} & H \\
 \Delta \searrow & & & & \nearrow m \\
 & H \otimes H & \xrightarrow{\text{id} \otimes S} & H \otimes H &
 \end{array}$$

## 2 Enriched Categories

### 2.1 Basic Definitions

**Definition 2.1.1 (Enriched Categories)** Let  $(\mathcal{S}, \times, I, \alpha, \lambda, \rho)$  be a monoidal category. Let  $\mathcal{C}$  be a category. We say that  $\mathcal{C}$  is enriched over  $\mathcal{S}$  if the following are true.

- For each  $C, D \in \mathcal{C}$ ,  $\text{Hom}_{\mathcal{C}}(C, D) \in \mathcal{S}$
- For each  $C \in \mathcal{C}$ , there is a morphism  $u_C : I \rightarrow \text{Hom}_{\mathcal{C}}(C, C)$  in  $\mathcal{S}$
- For each  $C, D, E \in \mathcal{C}$ , there is a morphism in  $\mathcal{S}$  given by

$$\circ : \text{Hom}_{\mathcal{C}}(D, E) \times \text{Hom}_{\mathcal{C}}(C, D) \rightarrow \text{Hom}_{\mathcal{C}}(C, E)$$

such that the following diagram commutes:

- Associativity:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(E, F) \times (\text{Hom}_{\mathcal{C}}(D, E) \times \text{Hom}_{\mathcal{C}}(C, D)) & \xrightarrow{1 \times \circ} & \text{Hom}_{\mathcal{C}}(E, F) \times \text{Hom}_{\mathcal{C}}(C, E) \\ \uparrow \alpha & & \downarrow \circ \\ (\text{Hom}_{\mathcal{C}}(E, F) \times \text{Hom}_{\mathcal{C}}(D, E)) \times \text{Hom}_{\mathcal{C}}(C, D) & \xrightarrow{\circ \times 1} & \text{Hom}_{\mathcal{C}}(D, F) \times \text{Hom}_{\mathcal{C}}(C, D) \\ & & \uparrow \circ \\ & & \text{Hom}_{\mathcal{C}}(C, F) \end{array}$$

- Identity:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(C, D) \times I & \xrightarrow{\text{id} \times u} & \text{Hom}_{\mathcal{C}}(C, D) \times \text{Hom}_{\mathcal{C}}(C, C) \\ \searrow \cong & & \downarrow \circ \\ & & \text{Hom}_{\mathcal{C}}(C, D) \end{array} \quad \begin{array}{ccc} \text{Hom}_{\mathcal{C}}(D, D) \times \text{Hom}_{\mathcal{C}}(C, D) & \xleftarrow{u \times \text{id}} & I \times \text{Hom}_{\mathcal{C}}(C, D) \\ \downarrow \circ & & \swarrow \cong \\ \text{Hom}_{\mathcal{C}}(C, D) & & \end{array}$$

**Proposition 2.1.2** Let  $\mathcal{S}$  be a symmetric monoidal closed category. Then  $\mathcal{S}$  is enriched over itself.

### 2.2 Enriched Functors and Natural Transformations

**Definition 2.2.1** Let  $(\mathcal{S}, \otimes, I)$  be a monoidal category. Let  $\mathcal{C}$  and  $\mathcal{D}$  be categories enriched over  $\mathcal{S}$ . An  $\mathcal{S}$ -functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  consists of the following data.

- $F : \mathcal{C} \rightarrow \mathcal{D}$  is a functor.
- The assignment  $F : \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$  is a morphism in  $\mathcal{S}$ .

Moreover, the following diagrams are commutative:

- Composition axiom:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(B, C) \otimes \text{Hom}_{\mathcal{C}}(A, B) & \xrightarrow{\circ} & \text{Hom}_{\mathcal{C}}(A, C) \\ F \otimes F \downarrow & & \downarrow F \\ \text{Hom}_{\mathcal{D}}(F(B), F(C)) \otimes \text{Hom}_{\mathcal{D}}(F(A), F(B)) & \xrightarrow{\circ} & \text{Hom}_{\mathcal{D}}(F(A), F(C)) \end{array}$$

- Unit axiom:

$$\begin{array}{ccc} I & \xrightarrow{u} & \text{Hom}_{\mathcal{C}}(A, A) \\ & \searrow u & \downarrow F \\ & & \text{Hom}_{\mathcal{D}}(F(A), F(A)) \end{array}$$

**Definition 2.2.2 (Enriched Natural Transformations)** Let  $(\mathcal{S}, \otimes, I)$  be a monoidal category. Let  $\mathcal{C}, \mathcal{D}$  be  $\mathcal{S}$ -categories. Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be two  $\mathcal{S}$ -functors. An  $\mathcal{S}$ -natural transformation  $\alpha : F \Rightarrow G$  consists of the following data. For each  $C \in \mathcal{C}$ , a morphism

$$\alpha_C : I \rightarrow \text{Hom}_{\mathcal{D}}(F(C), G(C))$$

such that the following diagram commutes in  $\mathcal{S}$ :

$$\begin{array}{ccccc}
 & & \text{Hom}_{\mathcal{C}}(A, B) & & \\
 & \swarrow \rho^{-1} & & \searrow \lambda^{-1} & \\
 \text{Hom}_{\mathcal{C}}(A, B) \otimes I & & & & I \otimes \text{Hom}_{\mathcal{C}}(A, B) \\
 \downarrow G \otimes \alpha_A & & & & \downarrow \alpha_B \otimes F \\
 \text{Hom}_{\mathcal{D}}(G(A), G(B)) \otimes \text{Hom}_{\mathcal{D}}(F(A), G(A)) & & & & \text{Hom}_{\mathcal{D}}(F(B), G(B)) \otimes \text{Hom}_{\mathcal{D}}(F(A), F(B)) \\
 & \searrow \circ & & \swarrow \circ & \\
 & \text{Hom}_{\mathcal{D}}(F(A), G(B)) & & & 
 \end{array}$$

**Definition 2.2.3 (Equivalence of Enriched Categories)** Let  $\mathcal{S}$  be a monoidal category. Let  $\mathcal{C}, \mathcal{D}$  be  $\mathcal{S}$ -categories. We say that  $\mathcal{C}$  and  $\mathcal{D}$  are  $\mathcal{S}$ -equivalent if there exists  $\mathcal{S}$ -functors  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{C}$  such that

$$\lambda : 1 \xrightarrow{\cong} F \circ G \quad \text{and} \quad \eta : G \circ F \xrightarrow{\cong} 1$$

are  $\mathcal{S}$ -isomorphisms.

TBA: Fully faithful essentially surjective.

Fully faithful implies isomorphism of Hom sets in  $\mathcal{S}$ .